Hooke’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect spring displacement with restoring force.

SUCCESS CRITERIA - I can...

1. Measure displacement from the natural length.
2. Use spring constant in N/m.
3. Point the spring force back toward equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “spring constant” and “displacement” to explain your answer. Sentence starters are optional.

spring constant: how much an ideal spring’s force changes per stretch

displacement: change from initial position to final position, with direction

equilibrium: zero net force and zero acceleration

01 Warm-up: bring back an earlier idea

A wheel rolls down a ramp, losing **15 J** gravitational energy. Its center-of-mass kinetic energy gain is **12 J** with negligible losses. What energy term is missing?

Word help: negligible = small enough to leave out of this model

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_{spring} = -kx$; $|F_{spring}| = k|x|$.
x is displacement from natural length; k is spring constant. The force restores the spring, within its linear range.

Worked reasoning: A spring with k=**100 N/m** stretches **0.03 m**, giving force magnitude kx=**3 N**. Its force on the attached object points toward equilibrium.

Your turn: Sketch equilibrium and stretched positions, then **draw the restoring-force arrow**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

A spring is stretched right.
1. **Draw the force it exerts** on the attached block.
2. **What changes** if compressed left?

Sentence starter: In my work, I used … to show … . I labeled it … .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: 1 2 3

04 Apply the physics

You test an ideal spring with k = **100 N/m**.
1. **Calculate restoring-force magnitudes** for stretches of **0.03 m** and **0.06 m**.
2. **Label both displacements** and **show the force direction relative** to equilibrium.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says k doubles when the stretch doubles. **Explain which quantity actually doubles** for the same ideal spring.

Word help: ideal = using stated simplifying conditions

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **5 N** force acts forward through **7 m**. **Find work** and **show the force**/displacement directions. **Label force** and displacement, including their relative direction.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

07 Exit check

k=**200 N/m** and compression=**0.04 m**. **Find force size** and **describe direction**. **Mark the displacement** from equilibrium and **draw the restoring-force direction**.

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two springs have k=**50** and **100 N/m**. **Find stretches** that both give **4 N**, and **compare their stiffness**. **Mark the displacement** from equilibrium and **draw the restoring-force direction**.

Sentence starter: Both … . They differ in … because … .



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3